Deep Learning-Based DOA Estimation

Shilian Zheng, Zhuang Yang, Weiguo Shen, Luxin Zhang, Jiawei Zhu, Zhijin Zhao, and Xiaoniu Yang

***Abstract*—Direction-of-arrival (DOA) estimation is a vital research topic in array signal processing, with extensive applications in many fields. In recent years, deep learning has been applied to DOA estimation to improve the performance. However, most existing deep learning-based DOA estimation methods extract DOA information from the covariance matrix (CM) input. In this paper, we introduce a novel deep learning-based DOA estimation scheme that utilizes the raw in-phase (I) and quadrature (Q) components of the signal as the input. We formulate the problem as single-label classification and multi-label classification based on the number of signal sources. We design a convolutional neural network to solve the problems and to adapt to different number of snapshots. We also propose a deep learning regression-based method to overcome the limitations of classify-based methods in dealing with off-grid angles. We conduct extensive experiments with simulations and over-the-air collected signals to analyze the performance of the proposed method in various scenarios including different SNRs, additive generalized Gaussian noise (AGGN) and extreme multi-source DOA estimation. Results demonstrate that our proposed method outperforms the existing deep learning-based DOA estimation methods.**

***Index Terms*—Direction-of-arrival (DOA) estimation, deep learning, residual neural network, classification, regression.**

I. Introduction

ARRAY signal processing employs a sensor array consisting of multiple antennas to receive and process signals, enabling the extraction of useful information from them. Direction-of-arrival (DOA) estimation, a fundamental research component of array signal processing, has found widespread applications in various domains such as radar, sonar, communications, and speech processing [1], [2].

The beamforming method, an elementary technique for DOA estimation, extends the time-domain Discrete Fourier Transform (DFT) to the spatial domain. However, this method becomes infeasible when the beamwidth is greater than the spacing between the sources [3]. Subsequent to the beamforming method, various approaches based on spatial spectral estimation and maximum likelihood [4] were developed to estimate DOA. Examples of such methods include the MUltiple SIgnal Classification (MUSIC) [5] algorithm, the Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) [6] and their variants. The MUSIC method is a popular DOA estimation method that constructs spatial spectral peaks by separating the signal subspace and noise subspace which are orthogonal to each other. The orientation can be estimated from the corresponding spectral peaks by using the orthogonal property of the two subspaces. The CS-MUSIC [7] algorithm is a variant of the MUSIC algorithm that uses hybrid parametric minimization in conjunction with MUSIC for DOA estimation, which overcomes the coherence limitation. The ESPRIT method is another commonly used DOA estimation method that divides the array into two identical subarrays in structure. The corresponding array elements of the two subarrays are divided into pairs of forms and have the same translation distance between each pair. This approach ensures that the incident angle differs by only one rotationally invariant factor on the two subarrays, and the DOA of the incident signal can be obtained by solving for a generalized eigenvalue. Unitary ESPRIT [8] is a variant of ESPRIT, which uses the centrosymmetric property of isometric linear matrices to convert complex covariance matrices into real symmetric matrix. This transformation reduces the computational effort and provides higher accuracy in DOA estimation.

Another category of DOA estimation methods is based on sparse representation. In [9], the authors proposed a sparse representation DOA estimation method based on sensor measurements to enhance sparsity and improve the accuracy of estimation using the $\ell_1 - norm$ penalty. In [10], compressive sensing was used in DOA estimation. The compressive sensing method estimates DOA by recovering the sparse signal vector [11]. In [12], the authors proposed a gridless DOA estimation method based on a well-established covariance fitting criterion. This method continuously estimates the parameters over some time, resulting in a sparse parameter estimate that does not require the angular space to be partitioned into small blocks, thus avoiding discretization.

In recent years, deep learning [13] has been extensively studied in the fields of computer vision [14] and natural language processing [15]. It has also been applied in the field of radio signal processing, including signal detection [16], [17], modulation recognition [18], [19], [20], [21], channel estimation [22], and information recovery [23]. Similarly, in DOA

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estimation, deep learning-based methods have emerged as an alternative to traditional DOA estimation methods [24], [25].

Most deep learning-based DOA estimation methods tackled the problem using a classification model [24], [25], [26], [27], [28], [29]. In [24], the authors attempted to solve the DOA estimation problem using a deep neural network (DNN) with the lower triangular part of the Covariance Matrix (CM) as input for two signal sources. The neural network was trained in parallel using two different training sets, with the first set generating two random signal angles and the second set restricting two signal angles to 1° intervals. A successful prediction was considered when at least one of the two DNNs correctly estimated the DOAs. However, there are no established methods for selecting the appropriate DNNs for a given problem. In [25], a DNN was utilized to estimate the DOAs of two sources. The DNN in this work had four intermediate layers, each with nine different choices for the number of units. A total of 36 DNNs were trained, and the one with the highest accuracy rate was selected for DOA estimation. In [26], a deep convolutional network was used for DOA estimation with sparse prior conditions. The method aims to recover the spatial spectrum by measuring the obtained covariance matrix data, using a Convolutional Neural Network (CNN) to learn the spectral features. Different activation functions were utilized in the hidden layer to evaluate the model's performance. This method takes into account the spatial sparsity of the incident signal, which is an improvement over previous methods. By incorporating a priori information, the design and training of the network are made more efficient. The authors of [27] proposed a deep learning framework called CNN-based DeepMUSIC, which utilizes a divide-and-conquer approach by splitting the DOA region into multiple sub-regions. For each sub-region, a corresponding CNN is designed to estimate the DOA. The authors in [28] addressed the problem of DOA estimation in low SNR conditions by utilizing multichannel data as input, where the real part, imaginary part, and phase of the covariance matrix are used as three separate channels. A 2-dimensional convolutional layer is used to extract relevant features for more accurate DOA estimation. In [29], a novel online DOA estimation approach was proposed, which utilizes a combination of CNN and Long Short-Term Memory (LSTM) network.

Deep learning-based DOA estimation methods can also be developed by combining encoders and neural networks. In [30], a DOA estimation method was introduced using a combination of self-encoder and fully connected network structure, namely Multilayer Perceptron (MLP). This approach uses a multi-layer self-encoder to divide the signal into multiple spatial sub-regions, similar to a spatial filter and employs a classifier in each subregion for spatial spectrum estimation. The DNN model is trained using array-specific outputs, ensuring robustness against various defects. In [31], a residual network-based model is proposed to address defects in antenna arrays. The model determines the presence or absence of signals in corresponding subregions using a spatial classification network, followed by classifying the outputs of different subregions using the Residual Network (ResNet). This approach leads to improved accuracy in DOA estimation. In [32], a compressed data-based neural network model was proposed for accurate DOA estimation using a joint training of encoder and classifier. The model uses encoded data from each subarray as input to the classifier, allowing for accurate DOA estimation even with limited data.

Deep learning-based DOA estimation methods have demonstrated superior performance compared to traditional approaches. However, most of these methods relies on extracting information from the CM, which is derived from the original IQ signal. To simplify the process and eliminate the need to compute the covariance matrix, we propose to use the raw in-phase (I) and quadrature (Q) of the signal directly as input to the deep neural network for DOA estimation. This method exploits the powerful learning ability of deep learning on raw data to extract DOA information directly from the raw signal. We investigate the problems of estimating the DOA for both single source and multi-signal sources. We propose both classification-based and regression-based methods to solve the problems. We conduct extensive experiments under different conditions to evaluate the performance of the proposed method. We also compare it with the MUSIC method and three representative DOA estimation methods based on deep learning. The CNN-based method [28] and the MLP-based method [30] use covariance as input, and the sparse prior-based method [26] uses the spectral proxy (SP) calculated from CM as input. For simplicity, in the rest of the paper, we will refer to them as CM-CNN [28], CM-MLP [30], and SP-CNN [26].

The main contributions of this paper are as follows.

- We propose a novel DOA estimation method based on raw IQ input, namely IQ-ResNet, where the real and imaginary parts of the received signal are used as direct inputs to the ResNet. Using raw IQ data as input, information can be extracted directly to improve the performance of DOA estimation. Results show that the proposed IQ-ResNet is superior to the existing methods for obtaining DOA information from CM in both single-source DOA estimation and multi-source DOA estimation.
- We introduce two DOA estimation methods: classification-based method and regression-based method. For the angles present in the training set, the classification-based method generally has better performance. However, for the angles not present in the training set, the regression-based method generally has better performance. This highlights the flexibility and adaptability of regression-based approaches when dealing with angles that do not appear in the training grid.
- We design a ResNet structure to adapt to different snapshot numbers. Specifically, we extract the input signal dimension to one through the global average pooling layer, so that no matter how the signal length changes in the inference stage, the trained network can adapt without any adjustment. The simulation results show that the DOA estimation performance of the network trained in a certain number of snapshots can approximate the performance of the re-trained network, which shows its superior adaptability.

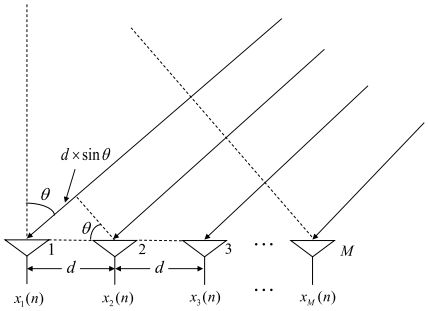


Fig. 1. ULA structure.

- We conduct extensive simulations to analyze the performance of the proposed method in various scenarios including different SNRs, Additive Generalized Gaussian Noise (AGGN), different sources, random angles, and extreme multi-source DOA estimation. The results show that in all these scenarios, the proposed IQ-ResNet is superior to the existing methods. Specifically, the model trained in AWGN environment can also adapt to DOA estimation in AGGN environment, which shows that it has strong generalization ability. Furthermore, the model trained in a two-source environment can be adapted to DOA estimation in different source environments, which shows its strong adaptability. More importantly, in the extreme multi-signal DOA estimation scenario, that is, estimating the DOAs of 7 signal sources with 8 receiving antennas, the proposed IQ-ResNet can estimate the DOAs of all 7 signals with a satisfactory accuracy, while the existing methods fail to estimate all of the DOAs at the same time.
- We build a practical test system and collect signals over-the-air in an anechoic chamber, and evaluate the performance of the algorithms with the collected signals. Experimental results show that the performance of the proposed IQ-ResNet is still better than that of the existing methods, which further illustrates the superiority of the proposed method.

The rest of this paper is organized as follows. In Section II, we discuss the signal model for DOA estimation. In Sections III and IV, the DOA estimation methods proposed in this paper are described in detail. In Sections V and VI, we show the simulation results and experimental results respectively. Finally, Section VII concludes the paper.

II. Signal Model

DOA estimation is typically accomplished through antenna array reception. Assume that the number of uniform linear array (ULA) elements in space is M and the array element spacing is d. The ULA structure is shown in Fig. 1. There are L-independent narrowband sources incident on the array elements in the directions $\theta_1, \theta_2, \ldots, \theta_L$. With the first array element as the reference, each narrowband signal is represented as $s_1(n), \ldots, s_L(n)$, then the signal received by the m-th array element is

$$x_m(n) = \sum_{i=1}^{L} s_i(n) e^{-j\frac{2\pi}{\lambda}(m-1)d\sin\theta_i} + \delta_m(n), \tag{1}$$

where $\lambda = c/f$ is the wavelength at carrier frequency f, c is the speed of light, $\delta_m(n)$ denotes the noise on the m-th array element. The received signal vector can be expressed as

$$\mathbf{x}(n) = \mathbf{A}(\theta)\mathbf{s}(n) + \boldsymbol{\delta}(n), \tag{2}$$

where $\mathbf{s}(n) = [s_1(n), \ldots, s_L(n)]^{\mathrm{T}} \in \mathbb{C}^{L\times 1}$ denotes the signal vector, $\boldsymbol{\delta}(n) = [\delta_1(n), \ldots, \delta_M(n)]^{\mathrm{T}} \in \mathbb{C}^{M\times 1}$ denotes the noise vector, $[\bullet]^{\mathrm{T}}$denotes transposition, $\mathbf{A}(\theta) = [\mathbf{a}(\theta_1), \ldots, \mathbf{a}(\theta_L)] \in \mathbb{C}^{M\times L}$ is the direction matrix, where $\mathbf{a}(\theta_\mathbf{i})$ is the direction vector of the i-th signal which can be expressed as

$$\mathbf{a}(\theta_i) = \left[1, e^{-j\frac{2\pi}{\lambda}d\sin\theta_i}, \ldots, e^{-j\frac{2\pi}{\lambda}(M-1)d\sin\theta_i}\right]^{\mathrm{T}}. \tag{3}$$

By collecting T time-sampling snapshots, Eq. (2) can be written as

$$\mathbf{X} = \mathbf{AS} + \boldsymbol{\delta}, \tag{4}$$

where $\mathbf{X} = [\mathbf{x}(0), \ldots, \mathbf{x}\ (T-1)]$, $\mathbf{A} = \mathbf{A}(\theta)$, $\mathbf{S} = [\mathbf{s}(0), \ldots, \mathbf{s}(T-1)]$, $\boldsymbol{\delta} = [\boldsymbol{\delta}(0), \ldots, \boldsymbol{\delta}(T-1)]$. All we have to do is to estimate the DOA of each narrowband signal from the received signal $\mathbf{X}$. When only one narrowband signal is incident on the array, i.e., $L = 1$, we refer to it as a single-source DOA estimation problem. At this time, $\mathbf{A}(\theta) = [\mathbf{a}(\theta_1)] \in \mathbb{C}^{M\times 1}$, $\mathbf{s}(n) = [\mathbf{s}_1(n)]^{\mathrm{T}} \in \mathbb{C}^{1\times 1}$. When multiple narrowband signals are incident on the array simultaneously, i.e., $L > 1$, we refer to it as a multi-source DOA estimation problem.

The traditional DOA estimation method mainly uses the covariance matrix for estimation. The array covariance matrix, also known as the array autocorrelation matrix can be expressed as

$$\mathbf{R}_{xx} = \mathrm{E}\left\{\mathbf{x}(n)\mathbf{x}^{\mathrm{H}}(n)\right\} = \mathbf{APA}^{\mathrm{H}} + \mathbf{U}, \tag{5}$$

where $\mathbf{P} = \mathrm{E}\{\mathbf{s}(n)\mathbf{s}^{\mathrm{H}}(n)\}$ and $\mathrm{E}\{\bullet\}$ denotes the expectation. Assume that each noise is independently and identically distributed Additive White Gaussian Noise (AWGN), then

$$\mathbf{U} = \mathrm{E}\left\{\boldsymbol{\delta}(n)\boldsymbol{\delta}^{\mathrm{H}}(n)\right\} = \sigma^2\mathbf{I}, \tag{6}$$

where σ^2 is the noise power, $\mathbf{I}$ is the unit matrix and $[\bullet]^{\mathrm{H}}$ is the conjugate transpose. In practical applications, the autocorrelation matrix is agnostic, and the sampling covariance matrix $\tilde{\mathbf{R}}_{xx}$ is generally obtained using T sampling snapshot estimation:

$$\tilde{\mathbf{R}}_{xx} = \frac{1}{T}\sum_{n=0}^{T-1}\mathbf{x}(n)\mathbf{x}^{\mathrm{H}}(n). \tag{7}$$

III. Proposed Method With IQ Input

A. Problem Formulation

DOA estimation is essentially a regression problem. In order to solve this problem by using classification, we need to discretize the DOA range, and the discretized set is represented as

$$\mathbf{Q} = \{-Q_{\max}, \ldots, -Q_{\alpha}, 0, Q_{\alpha}, \ldots, Q_{\max}\}, \tag{8}$$

where $\mathbf{Q}$ is the set of all possible directions of arrival of the signal, $Q_{\max}$ is the maximum direction of arrival, $-Q_{\max}$ is the minimum direction of arrival, and Q_{α} is the resolution of the DOA estimate. In this way, we can consider DOA estimation as a multi-category classification problem, where the number of categories is

$$D = (Q_{\max} + Q_{\max})/Q_{\alpha}. \tag{9}$$

In this paper, we use deep learning to tackle the DOA estimation problem. Specifically, we formulate the single-source DOA estimation as a single-label multi-classification problem, where there is only one label to be predicted. It can be represented as

$$Q = \arg\max_{q \in \mathbf{Q}} \Pr(Q = q|\mathbf{X}), \tag{10}$$

where $\Pr(\bullet)$ denotes the probability and Q is the predicted DOA. On the other hand, we treat the multi-signal source DOA estimation as a multi-label multi-classification problem, where there are multiple labels to be predicted. Similarly, this problem can be represented as

$$Q_i = \arg\max_{q \in \mathbf{Q}} \Pr(Q_i = q|\mathbf{X}), i = 1, 2, \ldots, L, \tag{11}$$

where Q_i is the predicted DOA of the i-th signal source.

B. Raw IQ Input

As previously mentioned, most current deep learning-based DOA estimation methods utilize the covariance matrix as input. In contrast, we propose a novel approach of directly feeding the IQ component of the signal into the neural network for learning. Specifically, we first extract the real and imaginary parts of the received signal of each array element:

$$R_m(n) = \mathrm{Re}(x_m(n)), I_m(n) = \mathrm{Im}(x_m(n)), \tag{12}$$

where $R_m(n)$ and $I_m(n)$ are the I component and Q component of the signal $x_m(n)$, respectively, $\mathrm{Re}(x_m(n))$ and $\mathrm{Im}(x_m(n))$ denote the computation of the real and imaginary of $x_m(n)$, respectively. We then splice them together to form a matrix as

$$\mathcal{M}_{\mathrm{IQ}} = \begin{bmatrix} R_1(0) & R_1(1) & \cdots & R_1(T-1) \\ R_2(0) & R_2(1) & \cdots & R_2(T-1) \\ \cdots & \cdots & \cdots & \cdots \\ R_M(0) & R_M(1) & \cdots & R_M(T-1) \\ I_1(0) & I_1(1) & \cdots & I_1(T-1) \\ I_2(0) & I_2(1) & \cdots & I_2(T-1) \\ \cdots & \cdots & \cdots & \cdots \\ I_M(0) & I_M(1) & \cdots & I_M(T-1) \end{bmatrix}. \tag{13}$$

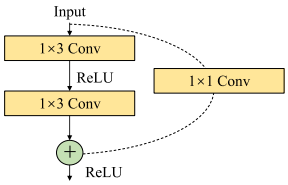


Fig. 2. The residual blocks used.

The dimension of the resulting matrix $\mathcal{M}_{\mathrm{IQ}}$ is $2M \times T$ and this matrix is used as the input to the subsequent neural network. This method bypasses the process of calculating the covariance matrix and extracts information directly from the original IQ signal, which is expected to yield better DOA estimation results.

C. Designed CNN

CNNs are multilayer nonlinear neural networks which were used for feature extraction of images at the beginning to solve the problem of high difficulty and low efficiency of manual feature extraction. Due to their powerful feature extraction capabilities, we leverage CNNs as classifiers for our problem. The key operation of the CNN is the convolutional layer, which receives the input and applies a series of convolutions with a set of kernels to extract relevant features. The general representation for the convolution layer is

$$x^{(l,p)} = \varphi\left(\sum_{i=1}^{m} a^{(l,p,i)} \otimes x^{(l-1,i)} + b^{(l,p)}\right), \tag{14}$$

where $x^{(l-1)}$ represents the input feature map of the l-th convolutional layer, and $x^{(l,p)}$ represents the output feature map of the same layer, $a^{(l,p,i)}$ is the l-th convolutional layer weight, $b^{(l,p)}$ is the l-th convolutional layer bias, and $\varphi(\bullet)$ is the activation function.

CNNs can indeed learn the features of the input, but blindly increasing the number of layers does not necessarily lead to improved accuracy. In the existing works [33], [34], [35], it has been pointed out that as the number of layers in a network increases, there is a likelihood of encountering issues such as gradient vanishing and gradient explosion. These challenges can impede the training of the model, leading to a situation where the accuracy of the model initially increases but subsequently experiences a rapid decline. Therefore, increasing the number of layers should be done carefully, and alternative techniques such as residual connections or skip connections should be considered to prevent the degradation of performance. The ResNet network [35] addresses the issue of gradient disappearance by incorporating direct mapping. Specifically, the input x of the residual module and the output $F(x)$ after two convolution layers are added together, and then the activation function $g(x)$ is applied to the summation result as the output of the residual module as

$$o = g(F(x) + x). \tag{15}$$

The residual block used in this paper passes the input x through a 1×1 convolution layer before adding it to $F(x)$, as shown in Fig. 2.

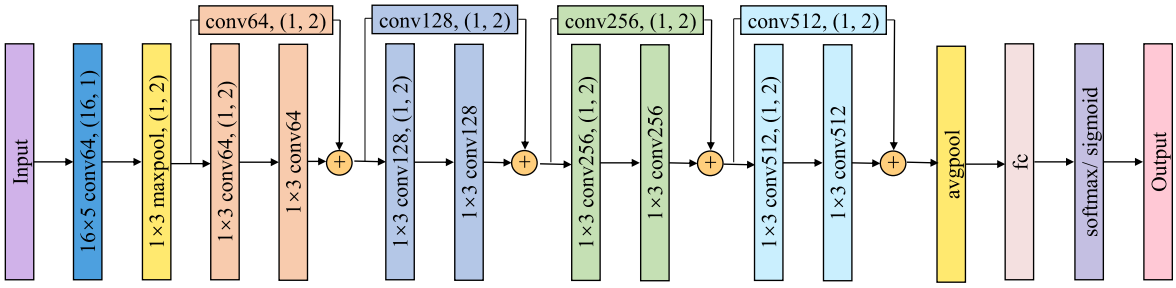


Fig. 3. The designed ResNet model applied to DOA estimation. "maxpool" stands for the maxpooling layer, "avgpool" stands for the average pooling layer, "fc" stands for the fully connected layer, and (a, b) means the convolution stride is (a, b).

It is only designed to make the direct mapping part and the part that goes through the convolution layer extract features of the same dimension. The residual block consists of two convolutional layers, a series of activation layers, and a normalization layer. For activation layers, Rectified Linear Unit (ReLU) activation function is used:

$$g(x) = \max(x, 0). \tag{16}$$

It is essentially a ramp function and introduces nonlinearity into the network, allowing it to approximate any nonlinear function. The ReLU activation function is computationally efficient and has been shown to perform well in many different applications. The normalization layer speeds up the training of the network by reducing the internal covariate bias, for example by normalizing all the input distributions to a standard normal distribution. The normalized layer used can be expressed as

$$z = \frac{y - \mathrm{E}(y)}{\sqrt{\mathrm{Var}(y) + \varepsilon}} \times \gamma + \beta, \tag{17}$$

where y is the input data, $\mathrm{E}(y)$ is the mean of the input data, and $\mathrm{Var}(y)$ is the variance of the input data, ε is a very small number which is used to avoid division by zero, γ and β are the learnable scale parameter and deviation parameter, respectively.

For DOA estimation, our designed CNN is mainly obtained based on the modification of basic ResNet18 [35] structure. It consists of a convolutional layer, four residual blocks, two pooling layers, and a fully connected layer. The detailed structure is shown in Fig. 3. It can be represented by the following nonlinear function:

$$f(x) = f_8(f_7(\cdots f_1(x))), \tag{18}$$

where $\{f_i(x)\}(i = 1, 2, \ldots, 8)$ are defined as follows.

- $f_1(x)$ denotes the first convolutional layer which is used to extract the signal features and transform the input matrix into a sequence for each channel. The number of convolution kernels is 64. The size of the convolution kernel is $2M \times 5$ and the step size is $2M \times 1$. With this configuration, correlation among array elements can be explored and learned. Batch-normalization and activation are also included in this layer.
- $f_2(x)$ denotes the maximum pooling layer. The pooling operation reduces the dimensionality of the feature map while retaining the most important information, thus improving the efficiency and effectiveness of the subsequent layers. By dividing the input data into smaller regions and performing operations on them, the pooling layer helps to capture local features and reduce the impact of small spatial shifts in the input data.
- $\{f_i(x)\}$ $(i = 3, 4, 5, 6)$ represent the four residual blocks, each of which with the structure shown in Fig. 2. In the two convolutional layers used to extract features in the residual block, the size of the first convolutional kernel is 1×3 with a step size of 1×2 and the size of the second convolutional kernel is 1×3 with a step size of 1×1. The four residual blocks have varying output channel numbers. The first has 64 channels, the second has 128 channels, the third has 256 channels, and the fourth has 512 channels.
- $f_7(x)$ denotes the averaging pooling layer. In order to reduce the feature map size to the greatest extent, we apply global averaging pooling to the data and output only one value per channel. This approach not only decreases the number of network parameters but also mitigates the risk of overfitting.
- $f_8(x)$ denotes the fully connected layer. The features extracted from the preceding convolution and pooling layers are weighted and combined. In the case of a classification problem, a fully connected layer takes the learned features and expands them into a one-dimensional vector, which is then mapped to labels in the sample space. The final fully connected layer consists of $D = 2Q_{\max}/Q_\alpha$ neurons, each representing an angle on the pre-defined grid.

D. Training and Inference

The output layer is located immediately after the fully connected layer at the end of the network structure. In classification problems, the output layer generates probabilities for each class. The commonly used output layers are softmax and sigmoid. The softmax function is generally used for single-label classification problems, while the sigmoid function is used for multi-label classification problems where there are multiple correct answers. In this paper, we use softmax and sigmoid for single-source DOA estimation problem and multi-source DOA estimation problem, respectively.

1) Single Source: In the case of DOA estimation for a single signal source, the softmax layer transforms the output

values of the fully connected layer into a probability distribution within the range [0, 1] by computing

$$\mathrm{softmax}(x_i) = \frac{\exp(x_i)}{\sum_{j=1}^{D} \exp(x_j)}, \tag{19}$$

where x_i is the output of the i-th neuron in the fully connected layer. The value on each neuron represents the confidence level of the presence of the signal in the corresponding direction, and the sum of the function output values is 1. In this paper, we employ cross-entropy as the loss function for single-source DOA estimation:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{B} \sum_{k=1}^{B} \sum_{d=1}^{D} p_{kd} \log(q_{kd}), \tag{20}$$

where B is the number of samples in the mini-batch, D is the number of categories, q_{kd} is the predicted probability of the k-th sample in the d-th category, and p_{kd} is the true label of the k-th sample in the d-th category.

In the inference stage, the label corresponding to the maximum confidence level is the predicted result, i.e.,

$$Q = q_i, i = \arg \max_{d \in \{1,\ldots,D\}} \mathrm{softmax}(x_d), \tag{21}$$

where q_i denotes the i-th element in the set $\mathbf{Q}$.

2) Multiple Sources: For multi-signal sources, we choose the sigmoid function instead of softmax due to the presence of multiple labels to be predicted. The sigmoid layer similarly restricts the output value to be between 0 and 1 by

$$\mathrm{sigmoid}(x_i) = \frac{1}{1 + \exp(-x_i)}. \tag{22}$$

This value represents the confidence that a signal source exists at that angle, rather than a true probability. In this case, we use the binary cross-entropy as the loss function for training multi-source DOA estimation:

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{B \times \mathrm{D}} \sum_{k=1}^{B} \sum_{d=1}^{D} [t_k(d) \log(o_k(d)) + (1 - t_k(d)) \log(1 - o_k(d))], \tag{23}$$

where $o_k(d)$ is the prediction result of the k-th sample belonging to the d-th category, and $t_k(d)$ is the true label of the k-th sample in the d-th category.

After the model is trained, in the inference stage, the test sample is fed into the trained model and the output confidence levels are rearranged from largest to smallest, and the labels corresponding to the top L confidence levels are taken as the predicted results of the L signal sources, i.e.,

$$Q_i = q_l, l \in \{l\} = \mathrm{topL}_{d \in \{1,\ldots,D\}} \mathrm{sigmoid}(x_d), \tag{24}$$

where topL denotes the calculation of top L values.

IV. Proposed Regression-Based Method

Classification-based DOA estimation methods have limitations in estimating signals outside a predefined grid set. In practical scenarios, the angle of the signal source can take any value, whereas the classification network can only classify it as the closest discrete angle within the grid set. The pre-set grid set limits the performance of DOA estimation based on the classification model. To address this issue, we propose a regression-based DOA estimation method. Unlike classification methods, the regression-based method directly outputs the predicted DOA instead of the class confidence. By utilizing a regression-based approach, we aim to overcome the limitations imposed by the grid set and provide more accurate and continuous DOA estimation, enhancing the overall performance and flexibility of the DOA estimation model.

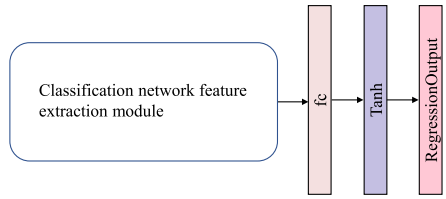


Fig. 4. Regression network applied to DOA estimation.

In the classification model, the output of the linear layer is passed through either a softmax layer or a sigmoid layer to generate the confidence levels associated with each category. However, in the regression model, the linear layer outputs L values. After the linear layer, the tanh layer limits the output to the range -1 to 1 by

$$\tanh(x_i) = \frac{e^{x_i} - e^{-x_i}}{e^{x_i} + e^{-x_i}}. \tag{25}$$

This step helps to normalize the output values. The labels utilized in the regression method represent the true DOA values. In order to promote better convergence of the model, these labels are narrowed down to a range of -1 to 1 when training. The remaining components of the model are consistent with the classification model, as shown in Fig. 4. By adapting the architecture in this manner, we can effectively train regression-based networks to estimate DOA. The introduction of the tanh layer and appropriate label scaling aids in achieving optimal convergence and accurate estimation within the specified DOA range.

For the regression method, Mean Squared Error (MSE) is used as the loss function of the model, which can be expressed as

$$\mathcal{L}_{\mathrm{MSE}} = -\frac{1}{B \times \mathrm{D}} \sum_{l=1}^{L} \sum_{i=1}^{B} \left(\theta_i^{(l)} - \tilde{\theta}_i^{(l)} \right)^2, \tag{26}$$

where $\theta_i^{(l)}$ is the true DOA of the l-th source of the i-th sample, and $\tilde{\theta}_i^{(l)}$ is the l-th predicted result of the i-th sample.

V. Simulation Results

In this section, we describe the simulation setup and the results in various scenarios. To verify the performance of the proposed method, we compare it with the methods of obtaining DOA information from covariance, i.e., CM-CNN, CM-MLP, SP-CNN and MUSIC.

A. Simulation Setup

1) Dataset: We use simulations to generate the datasets. First, binary bit streams are randomly generated and they are modulated using BPSK. A raised cosine filter with a roll-off factor of 0.5, a number of affected symbols of 6, and an oversampling rate of 8 are subsequently used for pulse shaping. The noise are added after taking T sampling points. After power normalization of the received signal, we separate its real and imaginary parts to obtain the I and Q components of the signal. The number of array elements M is set to 8 unless otherwise specified. Therefore, the size of each sample is $16 \times T$. We set the maximum DOA $Q_{\max} = 90^\circ$ and the resolution $Q_\alpha = 1^\circ$, defining a grid set $\mathbf{Q} = \{-90^\circ, -89^\circ, \ldots, 89^\circ, 90^\circ\}$ with a total of 181 angles. For the classification network, this means that the output of our network has 181 neurons with each neuron representing a specific DOA. Each neuron corresponds to a category of 1 if there is a signal and 0 if there is no signal. For the regression network, the label assigned to each signal corresponds to its true DOA. We here consider the DOA estimation in three cases: single-signal source, two signal sources and seven signal sources.

- Single source. When there is only one signal source, the 181 neurons outputted by the network have only one correct answer, which is the direction in which the signal exists. As mentioned earlier, we model single-source DOA estimation as a single-label multi-classification problem, where the label corresponds to the direction of the signal source. For the grid $\mathbf{Q} = \{-90^\circ, -89^\circ, \ldots, 89^\circ, 90^\circ\}$, we generate 50 samples for each direction at each SNR, resulting in a total of 9,050 samples for each SNR. We select SNRs in the range of -25 dB to 25 dB with an interval of 5 dB and use a snapshot of $T = 1024$. The training set comprises 99,550 samples, and the validation set is set up in the same way. Samples at different SNRs are combined as a training set and are trained using a single neural network, enhancing the model's ability to adapt to varying SNRs.
- Two Sources. When two sources are present simultaneously, the network's 181 neurons output two correct answers, and it is necessary to identify the two directions in which the signals are present from the 181 possible directions. As mentioned earlier, the DOA estimation problem in this case is modeled as a multi-label multi-classification problem. The presence or absence of a signal in each direction is treated as a binary classification problem, and the 181 neurons in the final fully connected layer are considered as 181 binary classifiers. Binary coding is used as the label in this case. The 181 grid points ranging from -90° to 90° correspond to a binary vector of size 181×1, where the presence of a signal source is denoted by 1 and its absence by 0. For example, label $e = [1, 1, 0, \ldots, 0]$ corresponds to the two directions $\{-90^\circ, -89^\circ\}$. The DOAs of the two signals are randomly selected from the grid $\mathbf{Q}$. This means that there is a total of $\mathrm{C}_{181}^2 = 16,290$ possible DOA combinations. For each DOA combination and each SNR, one sample with $T = 512$ is generated. The SNR ranges from -25 dB to 25 dB with a 5 dB interval. Thus, the training set has a total of $11 \times 16,290 = 179,190$ samples. The validation set is set up in the same way.
- Seven sources. For overdetermined DOA estimation with an array size of $M = 8$, we can estimate the DOA for up to seven sources. In this simulation, we focus on the DOA estimation of seven signal sources. The training set was generated using SNR values of $\{0, 5, 10, 15, 20\}$ dB and snapshots $T = 1024$. The DOAs of two of the signal sources are randomly selected from $\{10^\circ, 11^\circ, \ldots, 89^\circ, 90^\circ\}$, which means there is a total of $C_{81}^2 = 3240$ possible DOA combinations for these two signals. The rest of the five signal sources take the DOA randomly chosen from $[-90^\circ, -71^\circ]$, $[-70^\circ, -51^\circ]$, $[-50^\circ, -31^\circ]$, $[-30^\circ, -11^\circ]$, and $[-10^\circ, 9^\circ]$, respectively. In the training set, 10 samples are generated for each combination, resulting in a total of 162,000 samples. In the validation set, 3 samples are generated for each combination, resulting in a total of 48,600 samples.

2) Training Setting: For single-source DOA estimation, we use the Adam optimizer [36] with an initial learning rate of 0.001. The learning rate is changed every 10 epochs. The new learning rate is set 0.5 times of the previous value. A total of 30 epochs is used. The mini-batch size is set to 64. For multi-source DOA estimation, the same configuration is used except that the number of epochs is 40. The model with the highest accuracy on the validation set is selected as the final saved model. Note that the hyperparameter settings for training SP-CNN, CM-CNN and CM-MLP are the same as in the original paper [26], [28], [30]. All of the models are trained on a computer with Intel Core i7-9750H@2.60GHz CPU, 32 GB of running memory, and GeForce RTX2080 GPU. It's obvious that MUSIC does not need training process. When the MUSIC method performs spectrum search, the search range is -90° to 90°, and the step size is set 0.1°.

3) Performance Metrics: For single-source DOA estimation, in the test phase, we use RMSE to evaluate the performance of the model:

$$\mathrm{RMSE}_{\mathrm{single}} = \sqrt{\frac{1}{K}\sum\nolimits_{k=1}^{K}\left(\theta^{(k)} - \tilde{\theta}^{(k)}\right)^2}, \tag{27}$$

where K is the number of test samples, $[\theta^{(1)}, \ldots, \theta^{(K)}]$ are the true DOAs of the samples, and $[\tilde{\theta}^{(1)}, \ldots, \tilde{\theta}^{(K)}]$ are the predicted DOAs. We also evaluate the model performance using the average DOA, which is expressed as

$$\tilde{\theta} = \frac{1}{K}\sum_{k=1}^{K}\tilde{\theta}^{(k)}. \tag{28}$$

For multi-signal DOA estimation, we also use the RMSE to evaluate the model performance. But unlike single-source DOA estimation, here we have multiple real DOAs. The RMSE in this case is computed as

$$\mathrm{RMSE}_{\mathrm{multi}} = \sqrt{\frac{1}{KL}\sum\nolimits_{k=1}^{K}\sum\nolimits_{l=1}^{L}\left(\theta_l^{(k)} - \tilde{\theta}_l^{(k)}\right)^2}. \tag{29}$$

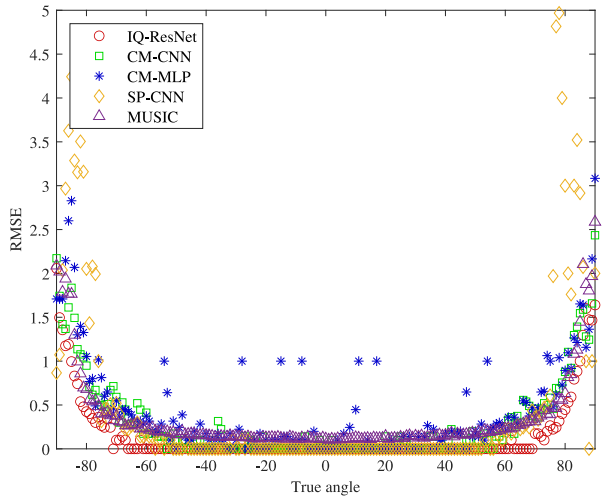


Fig. 5. Comparison of DOA estimation methods with single source.

Meanwhile, we use the RMSE of each signal source separately to evaluate the performance of the model. The RMSE of the l-th signal source is written as

$$\mathrm{RMSE}_{\mathrm{single}}(l) = \sqrt{\frac{1}{K}\sum\nolimits_{k=1}^{K}\left(\theta_l^{(k)} - \tilde{\theta}_l^{(k)}\right)^2}. \quad (30)$$

We also evaluate the model performance using the average DOA of each signal, which is expressed as

$$\tilde{\theta}_l = \frac{1}{K}\sum_{k=1}^{K}\tilde{\theta}_l^{(k)}. \quad (31)$$

Note that in computing these metrics, the true labels and the predicted results of the L signals are ranked from smallest to largest.

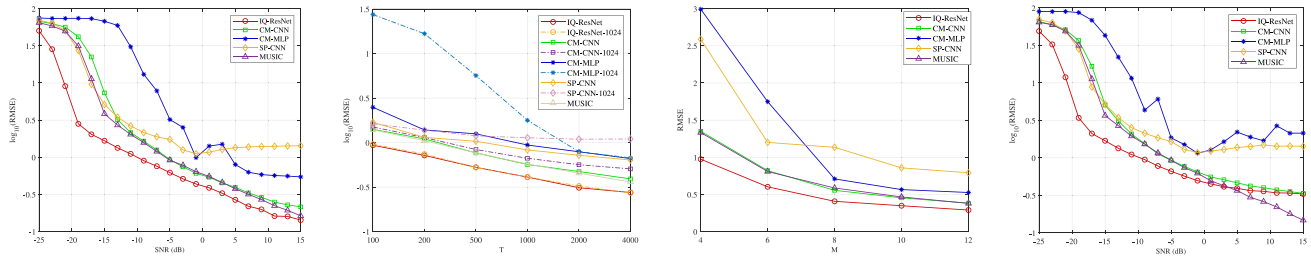
B. Results With Single Source

1) Comparison of Different Methods: We use samples with SNR $= 0$ dB (100 test samples for each DOA specifically) to compare the performance of our proposed IQ-ResNet with existing methods. Results are presented in Fig. 5. The proposed IQ-ResNet exhibits an RMSE close to zero in the range of $[-75°, 75°]$, with a gradual increase in the RMSE outside this interval. On the other hand, the CM-CNN achieves an RMSE close to zero only in the range of $[-40°, 40°]$, the SP-CNN achieves near zero RMSE in the range of $[-45°, 50°]$, and the CM-MLP shows an RMSE of zero only at a few angles, while the MUSIC method can not achieve zero RMSE at all angles. According to Cramér-Rao lower bound [37], we know that as the absolute angle increases, the estimation error will increase, which means that the RMSE in the angular edge region will be larger than the RMSE in the middle region. In the angular edge region, the IQ-ResNet method yields an RMSE between [0.47, 2.05], the CM-CNN method yields an RMSE between [1.04, 2.43], and the CM-MLP method produces an RMSE between [1.09, 3.08]. The SP-CNN method yields an RMSE between [0.3, 4.96], and the MUSIC method produces an RMSE between [0.39, 2.58]. These findings indicate that the proposed IQ-ResNet method outperforms the other four methods, thus demonstrating the superiority of the proposed approach.

2) Performance Under Different SNRs: We now analyze the performance of the methods by considering the SNR range of -25 dB to 15 dB with a 2 dB interval. There are 100 test samples for each angle at each SNR, resulting in a total of 18,100 test samples for each SNR. The simulation results are depicted in Fig. 6(a). The errors of all five methods decrease as the SNR increases. In the high SNR region, the decreasing trend of RMSE slows down significantly with the further increase of SNR. It's obvious that the proposed IQ-ResNet method achieves significantly smaller errors compared to the other four methods in the whole SNR range. Note that in these untrained SNRs, the proposed method can achieve low RMSE, which shows good generalization ability of the method.

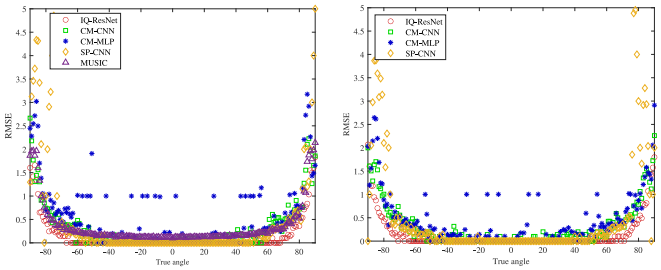
3) Adaptation to Different Snapshots: In this experiment, we aim to investigate the impact of varying the number of snapshots on the performance of the models. For simplicity, we consider estimating the DOA of signal with SNR $= 0$ dB. We consider six cases of snapshot numbers, namely 100, 200, 500, 1,000, 2,000, and 4,000. The RMSE is calculated for each snapshot by averaging over $K = 18{,}100$ test samples (i.e., 100 samples for each DOA). Two training approaches are used to evaluate the performance of the models. In the first approach, different models are trained for different snapshot numbers, with a specific model corresponding to a specific snapshot number. In the second approach, a single model is trained with snapshot number $T = 1024$, and DOA estimation is performed on data from other snapshots without retraining the model. The results are presented in Fig. 6(b). It can be observed that the RMSE of all five methods decreases as the number of snapshots increases. The IQ-ResNet method consistently outperforms the other four methods, with smaller RMSE across all snapshot numbers. The performance of the model trained under the $T = 1024$ snapshot is not as good as that of the retrained model. However, for the proposed IQ-ResNet, the difference in performance between the two models is very small. This suggests that the IQ-ResNet method is better adaptive to variations in the number of snapshots, and can provide more accurate DOA estimates even when the model is not retrained for a new number of snapshots.

4) Performance Under Different Number of Array Elements: In this experiment, we investigate the impact of different number of array elements on the performance of the DOA estimation model. We consider estimating the DOA of signal with SNR $= 0$ dB. We examine five scenarios with varying number of array elements, i.e., 4, 6, 8, 10, and 12. The models are retrained in each scenario. The RMSE is calculated for each number of array elements by averaging the results from $K = 18,100$ test samples (100 samples per DOA). The experimental results are shown in Fig. 6(c). It can be observed that the RMSE of the five methods decreases with the increase in the number of array elements. It is worth noting that the IQ-ResNet method presented in this paper shows the smallest RMSE in five scenarios, showing its superior performance in this respect.



(a) (b) (c) (d)

Fig. 6. Performance under different SNRs, different snapshots, different number of array elements and random angles. (a) Different SNRs, (b) different snapshots, (c) different number of array elements, and (d) random angles.



(a) (b)

Fig. 7. Performance of DOA estimation in AGGN. (a) AGGN-trained model and (b) AWGN-trained model.

5) Performance Under Random Angles: In this experiment, we deviate from using discrete angles for test and instead employ uniform random angles. Specifically, the test angle obeys a uniform distribution between $-90°$ and $90°$. We generate 10,000 samples with random angles for each SNR which is between -25 dB and 15 dB with a 2 dB interval. The results are presented in Fig. 6(d). It illustrates that in the low SNR region, our proposed IQ-ResNet exhibits lower RMSE and outperforms the other methods. In the high SNR region, its performance is better than other deep learning-based methods, but worse than traditional MUSIC. This is mainly because classification methods can only obtain directions on discrete grid points, and there will inevitably be estimation errors for angles on random off grid points. However, it should be pointed out that in high SNR regions, the estimation error is already very small. Taking into account these factors, the overall performance of our proposed method is still better than other methods.

6) Performance Under AGGN: Here, we investigate the performance of our model under AGGN [38]. The arbitrary location, inverse scale, and shape parameter of the AGGN are conventionally set to 0, 1, and 1.5, respectively. We first train the model with training samples with AGGN and test the performance using test samples with SNR $= 0$ dB. The results are presented in Fig. 7(a). The proposed IQ-ResNet exhibits an RMSE that is close to zero within the interval of $[-75°, 80°]$. On the other hand, the RMSE of the CM-CNN method is close to zero at most angles in the range of $[-40°, 40°]$, and the RMSE of the SP-CNN method is close to zero in the range of $[-40°, 50°]$, and the RMSE of the CM-MLP method is only close to zero at a few angles. However, the RMSE of the MUSIC method is not zero at all angles. Overall, the proposed IQ-ResNet outperforms the other four methods, thereby demonstrating the superiority of the proposed IQ-ResNet approach in AGGN scenario.

To illustrate the adaptability of the proposed method, we further evaluate the performance of the AWGN-trained model in AGGN scenario. The results are presented in Fig. 7(b). It is observed that the proposed IQ-ResNet can accurately estimate the DOA in the presence of AGGN even though it is trained with samples in the presence of AWGN. Furthermore, the RMSE of the IQ-ResNet method is consistently smaller than that of the other three methods over the entire DOA range. These results demonstrate the superior adaptability of the proposed IQ-ResNet method to new noise distributions.

C. Results With Two Sources

1) Comparison of Different Methods: We now consider two sources with an angular difference of $5°$ incident on the antenna array. The DOA of the first signal ranges from $-90°$ to $85°$, while the DOA of the second signal ranges from $-85°$ to $90°$, both in steps of $1°$. We use samples with SNR $= 0$ dB (100 test samples for each DOAs specifically) to compare the performance of our proposed IQ-ResNet with existing methods. The results are shown in Fig. 8. It is observed that the IQ-ResNet method achieves a maximum RMSE of 5.48 for the first signal source and 3.99 for the second signal source, while the CM-CNN method exhibits a maximum RMSE of 6.33 and 5.97 for the first and second signal sources, respectively. The CM-MLP method, on the other hand, has a maximum RMSE of 7.37 and 8.04 for the first and second signal sources, respectively. For the SP-CNN method, the maximum RMSE of the first signal is 20.9 and the maximum RMSE of the second signal is 17.8. For the MUSIC method the maximum RMSE of the first signal is 109.2 and the maximum RMSE of the second signal is 107.6. In the angular edge region, the RMSE of the CM-CNN method increases rapidly, and DOAs cannot be accurately estimated. Furthermore, the CM-MLP method and MUSIC method fails to accurately estimate both signal sources simultaneously. The proposed IQ-ResNet method has smaller RMSE and obtains the best performance.

2) Performance at Different SNRs: We now consider the performance of the methods in the SNR range of -25 dB to 25 dB with a 2 dB interval. We consider two sources with a $4°$ angular separation that are incident on the array, with the

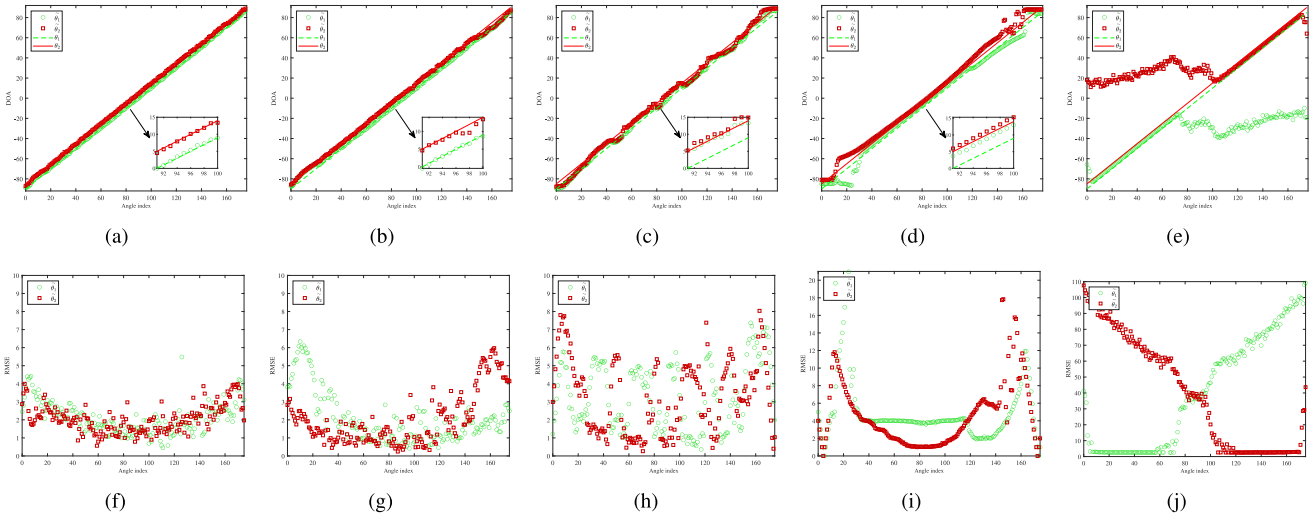


Fig. 8. Comparison of DOA estimation methods with two sources. (a) IQ-ResNet (mean), (b) CM-CNN (mean), (c) CM-MLP (mean), (d) SP-CNN (mean), (e) MUSIC (mean), (f) IQ-ResNet (RMSE), (g) CM-CNN (RMSE), (h) CM-MLP (RMSE), (i) SP-CNN (RMSE), and (j) MUSIC (RMSE).

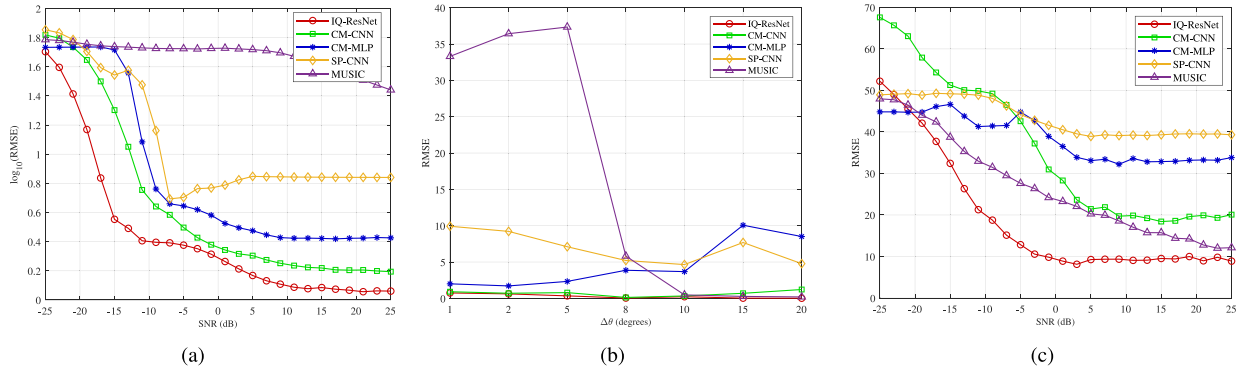


Fig. 9. Performance of DOA estimation of two sources under different SNRs, different angle separation and random angles. (a) Different SNRs, (b) different angle separation, and (c) random angles.

first source oriented from -90° to 86° and the second source oriented from -86° to 90°, both with a step of 1°. There are 100 test samples for each angle at each SNR, resulting in a total of 17,700 test samples for each SNR. Results are shown in Fig. 9(a). The RMSE of all five methods decrease as the SNR increases. At extremely low SNR (-25 dB), where noise has a significant impact, the RMSE of the five methods are similar. In the high SNR region, the RMSE of the four deep learning-based methods tends to stabilize as the SNR further increases. It's obvious that the proposed IQ-ResNet method achieves significantly smaller errors compared to the other four methods in the whole SNR range. It should be noted that although we do not train the model with all of the tested SNRs, the proposed method can still achieve low RMSE, which indicates that the method has good generalization ability.

3) Performance at Different Angle Separation: In this experiment, we evaluate the performance of the model under different angle separations. The antenna array receives signals from two sources with angle separations denoted as $\Delta\theta$. The first signal has an angle of 0°, and the second signal has an angle of $0^\circ + \Delta\theta$. The SNR $= 5$ dB. We consider seven values for $\Delta\theta$, i.e., 1°, 2°, 5°, 8°, 10°, 15°, and 20°. Each scenario is tested with 1000 samples. The results are shown in Fig. 9(b). Both the IQ-ResNet method and the CM-CNN method are able to accurately estimate the DOAs in all scenarios, but the IQ-ResNet method has a smaller RMSE. On the other hand, the CM-MLP method provides accurate DOA estimates only when the angular separation is small ($1^\circ \leq \Delta\theta \leq 5^\circ$). In contrast, the MUSIC method is only able to provide accurate DOA estimates when the angular separation is large ($10^\circ \leq \Delta\theta \leq 20^\circ$). The RMSE of the SP-CNN method is relatively high and does not vary much across all of the separations. These results clearly demonstrate the advantages of our proposed IQ-ResNet method over other methods. The method achieves low RMSE values in all angular separation scenarios, confirming its effectiveness in accurately estimating the DOAs.

4) Performance at Random Angles: In this experiment, the DOAs of the two sources obey a uniform distribution between -90° and 90°. The signals are generated with SNRs ranging from -25 dB to 25 dB, spaced at 2 dB intervals. To ensure statistical significance and robustness of the results, we generate 10,000 samples at each SNR level. The results are shown in Fig. 9(c). In the low SNR region, the RMSE of all five methods decreases as the SNR increases. However, in the high SNR region, the RMSE of existing deep learning-based methods tends to stabilize, showing limited improvement

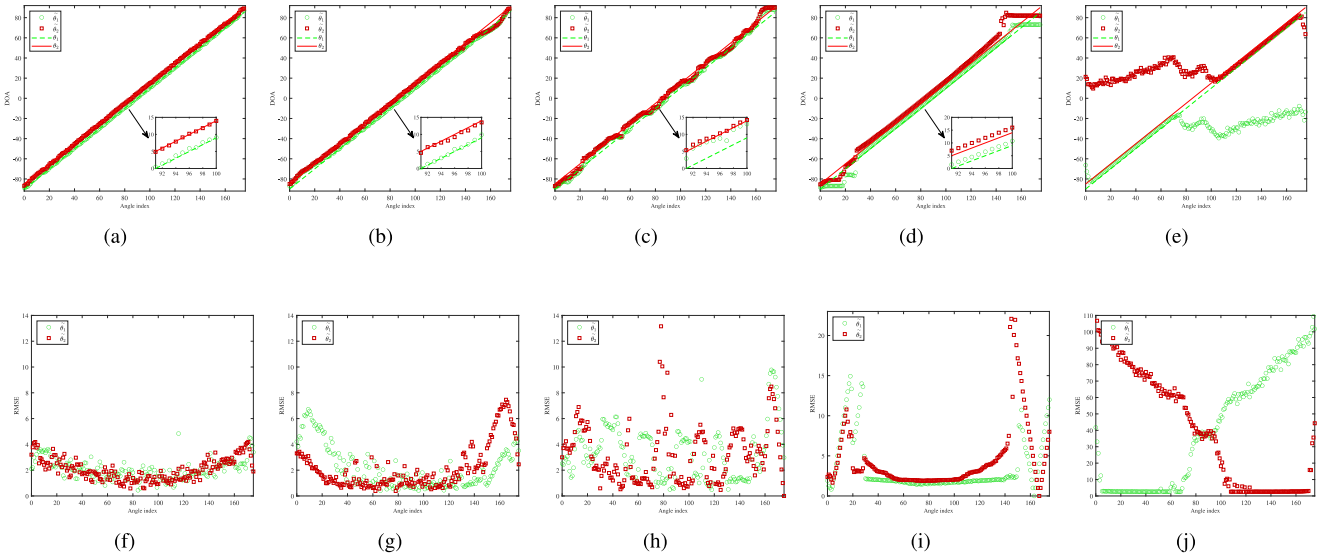


Fig. 10. DOA estimation performance in AGGN with two sources. (a) IQ-ResNet (mean), (b) CM-CNN (mean), (c) CM-MLP (mean), (d) SP-CNN (mean), (e) MUSIC (mean), (f) IQ-ResNet (RMSE), (g) CM-CNN (RMSE), (h) CM-MLP (RMSE), (i) SP-CNN (RMSE), and (j) MUSIC (RMSE).

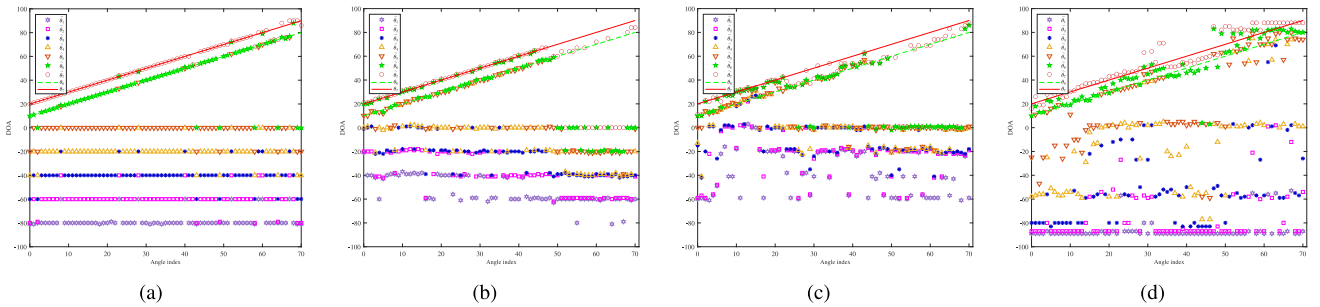


Fig. 11. DOA estimation of the seven signal sources. (a) IQ-ResNet, (b) CM-CNN, (c) CM-MLP and (d) SP-CNN.

with further increases in SNR. Throughout the entire SNR range, our proposed IQ-ResNet method consistently exhibits smaller RMSE compared to the other methods, highlighting its superiority and better performance in this task.

5) Performance Under AGGN: The performance of the model under AGGN is examined in this study using data with an SNR of 0 dB. Two signal sources with an angular difference of 5° are incident on the antenna array, with the first signal varying from -90° to 85° and the second signal varying from -85° to 90°, both with a step size of 1°. We take 100 test samples for each DOA. The results are presented in Fig. 10. In our observations, the IQ-ResNet method shows a maximum RMSE of 4.84 for the first signal and 4.19 for the second signal. In comparison, the CM-CNN method exhibits a maximum RMSE of 6.72 for the first signal and 7.45 for the second signal. The CM-MLP method, on the other hand, shows a maximum RMSE of 9.78 for the first signal and 13.16 for the second signal. For the SP-CNN method, the maximum RMSE of the first signal is 14.9 and a maximum RMSE of the second signal is 22.07. The MUSIC method has a maximum RMSE of 109.3 and 109.3 for the first signal and the second signal, respectively. In the edge regions of the angle range, the CM-CNN method can accurately estimate only one of the signal sources. The CM-MLP method and MUSIC method struggles to estimate both signal sources accurately, with only the first signal accurately estimated in certain angle regions and the second signal accurately estimated in other regions. The IQ-ResNet method exhibits less RMSE and superior performance compared to the other methods.

D. Results With Seven Sources

To assess the proposed method's performance in extreme situation, we simulate seven signal sources simultaneously incident on the antenna array. The incident angles of the first five sources are fixed at -80°, -60°, -40°, -20°, and 0°, respectively. The angle of incidence between the last two sources is fixed at a difference of 10°. The sixth signal source is incident at an angle ranging from 10° to 80°, and the seventh signal is incident at an angle ranging from 20° to 90° with a step of 1°. We use data with an SNR of 10 dB and 1 test sample for each DOA. The results are shown in Fig. 11. Note that the MUSIC method cannot give seven valid DOA estimates, so we do not give the corresponding results here. Our simulation results indicate that the IQ-ResNet method outperforms the other three methods in terms of accuracy and reliability for DOA estimation from seven sources. Specifically, the IQ-ResNet method can accurately estimate the DOAs of all seven sources in most of the angular regions. The CM-CNN method and SP-CNN method show

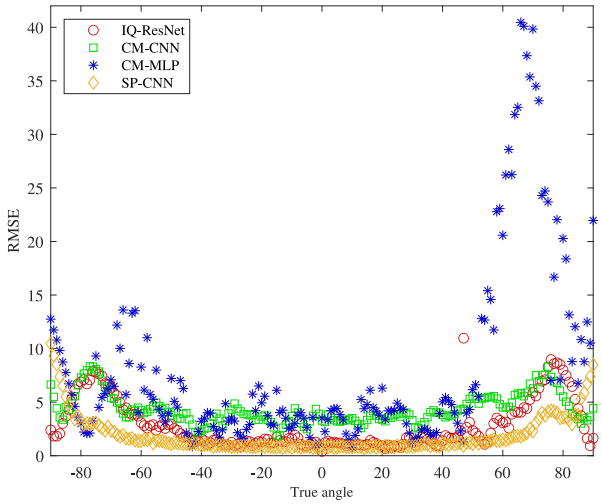


Fig. 12. Performance of a model trained with two sources in estimating DOA of a single source.

good performance for DOA estimation from up to six sources in the middle angular regions, and up to four sources in the final angular region. For the signal source with DOA of $-80°$, the CM-CNN method fails to accurately estimate its DOA in all cases. Similarly, the CM-MLP method shows limited success in DOA estimation from multiple sources. It can accurately estimate the DOAs of only three or four sources in certain angular regions, and fails to estimate the DOA of the signal source with DOA of $-80°$ accurately. Overall, the IQ-ResNet method demonstrates superior performance in DOA estimation of seven sources.

E. Adaptation to Different Number of Sources

In this experiment, we investigate the performance of a model trained with two-source signals in dealing with single-source and three-source signals. For the transfer performance analysis of a single signal source, we use SNR of -10 dB for the experiment, and each DOA contains 100 test samples. We compare the performance of the proposed IQ-ResNet method with existing methods. The results are shown in Fig. 12. In the range of $[-60°, 60°]$, the RMSE of our proposed IQ-ResNet method is small and ranges between [0.4, 3.5], the RMSE of the CM-CNN method ranges between [1.85, 5.57], and the RMSE of CM-MLP method ranges between [0.78, 23.06]. The SP-CNN method performs well in this case with an RMSE between [0.71, 1.6]. In the angular edge region, it is worth noting that the CM-MLP method has a larger error with a maximum RMSE of about 40.5.

In the three-source scenario, we consider three sources incident on the antenna array at the same time, where the angular difference between neighboring sources is fixed at $5°$. The DOA of the first signal ranges from $-90°$ to $80°$, the DOA of the second signal ranges from $-85°$ to $85°$, and the DOA of the third signal ranges from $-80°$ to $90°$, all in steps of $1°$. We use samples with an SNR of -10 dB for the experiment, and each DOA contains 100 test samples. The results are shown in Fig. 13. For our proposed IQ-ResNet method, the RMSE of the three signal sources is always below 6. Among them, the second source has the smallest RMSE, while the first and third sources have almost the same RMSE. For the CM-CNN and CM-MLP methods, the RMSE of the three sources is significantly higher in the angular edge region, indicating that these two methods cannot accurately estimate the DOAs of the three sources. Note that although the SP-CNN method performs well in single-source DOA estimation, it performs the worst in transfering to three sources. From this experiment we can observe that our proposed IQ-ResNet method performs the best and it exhibits good adaptability to both single-source and three-source scenarios. This adaptability further validates the effectiveness of our proposed method.

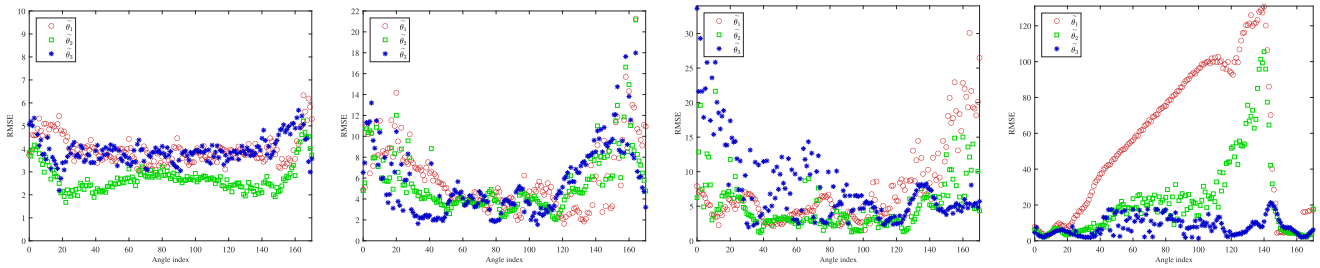
F. Performance of Regression-Based Method

Having discussed the performance of the classification-based DOA estimation methods, we now shift our focus to a regression-based approach for DOA estimation. We aim to compare the performance of regression-based and classification-based DOA estimation methods in both on-grid and off-grid cases. For simplicity, in the following we will refer to our proposed two methods using classification and regression as IQ-Classify and IQ-Regression, respectively.

1) Results With Single Source: We consider the performance of the proposed methods in both on-grid and off-grid scenarios. First, in the on-grid scenario, the DOA of the test samples ranges from $-90°$ to $90°$ with a step size of $1°$. We use samples with an SNR of 10 dB (specifically, 100 test samples per DOA) to compare the performance of our proposed IQ-Classify method and the IQ-Regression method. The experimental results are shown in Fig. 14(a). Within the range of $[-80°, 80°]$, the IQ-Classify method has better performance, with its RMSE between [0, 0.1]. The RMSE of the IQ-Regression method is between [0.19, 0.53]. At the edge of the angle region, the performance of the two methods is similar. Overall, for DOA estimation in the on-grid scenario, the IQ-Classify method performs better.

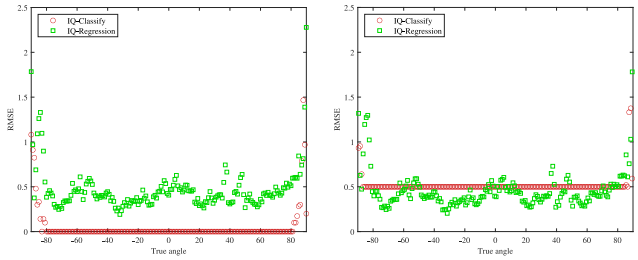
Next, we consider the off-grid scenario, where the DOA of the test samples ranges from $-89.5°$ to $89.5°$ in steps of $1°$. We use samples with an SNR of 10 dB (100 test samples for each DOA specifically). The experimental results are shown in Fig. 14(b). In the almost all angular region, the RMSE of the IQ-Classify method stabilizes at 0.5. However, most of the RMSE of the IQ-Regression method is below 0.5. The IQ-Classify method is limited by the grid set and cannot accurately predict the DOA in the off-grid scenario. In contrast, the IQ-Regression method is not limited by the grid set and shows better performance in this off-grid scenario.

2) Results With Two Sources: We focus on the regression-based method for DOA estimation performance in the presence of two sources. Here we consider two scenarios, i.e., on-grid and off-grid. First, in the on-grid scenario, we consider two sources with an angular difference of $5°$ incident on the antenna array. The DOA of the first source ranges from $-90°$ to $85°$, while the DOA of the second source ranges from $-85°$ to $90°$, both in steps of $1°$. We use samples with SNR $=$ 10 dB (specifically, 100 test samples per DOAs) to compare



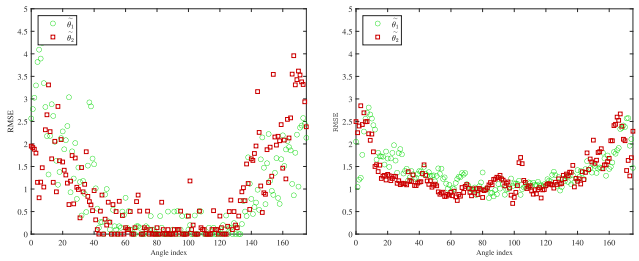
(a) (b) (c) (d)

Fig. 13. Performance of a model trained with two sources in estimating DOAs of three sources. (a) IQ-ResNet, (b) CM-CNN, (c) CM-MLP and (d) SP-CNN.



(a) (b)

Fig. 14. Comparison of the IQ-Classify and IQ-Regression methods with a single signal source. (a) On-grid scenario, (b) off-grid scenario.

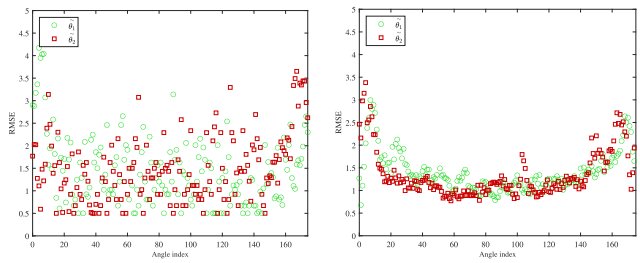


(a) (b)

Fig. 15. Comparison of IQ-Classify and IQ-Regression methods with two signal sources in the on-grid scenario. (a) IQ-Classify, (b) IQ-Regression.

the performance of the IQ-Classify method with that of the IQ-Regression method. The results are shown in Fig. 15. In the middle region of the angle, the IQ-Regression method has an RMSE in the range of [0.8, 1.45] for the first source, and an RMSE in the range of [0.68, 1.11] for the second source. In the angular edge region, the maximum RMSE is 2.85. On the contrary, the IQ-Classify method has an RMSE between [0, 0.8] for the first source and [0, 0.73] for the second source in the middle angular region. In the angular edge region, the maximum value of RMSE is 4.2. The IQ-Classify method has a smaller RMSE in most angular regions. In the on-grid scenario, the IQ-Classify method performs better.

Next, we consider the off-grid scenario where there are two sources with an angle difference of 5° incident on the antenna array. The DOA of the first signal ranges from -89.5° to 84.5°, while the DOA of the second signal ranges from -84.5° to 89.5°, both in steps of 1°. We use samples with SNR $=$ 10 dB (specifically, 100 test samples per DOAs) to compare the performance of the IQ-Classify method with that



(a) (b)

Fig. 16. Comparison of IQ-Classify and IQ-Regression methods with two signal sources in the off-grid scenario. (a) IQ-Classify, (b) IQ-Regression.

of the IQ-Regression method. The results are shown in Fig. 16. The IQ-Regression method has RMSE in the range of [0.67, 1.49] and [0.77, 1.79] for the first and second sources in the angular middle region. In the angular edge region, the maximum RMSE is 3.38. The IQ-Classify method has RMSE in the range of [0.5, 2.54] and [0.5, 3.2] for the first and second sources in the angular middle region. In the angular edge region, the maximum RMSE is 4.17. The RMSE of the IQ-Classify method fluctuates a lot, while the RMSE of the IQ-Regression method is relatively smooth. These results show that the IQ-Regression method exhibits better performance in estimating off-grid signals.

These results show that the classification-based approach exhibits excellent performance in the on-grid scenario, while the regression-based approach is more effective in the off-grid scenario.

G. Ablation Study and Complexity Analysis

1) Ablation Study: To further illustrate the advantages of the proposed method, we modify the network structure of the CM-CNN method to a similar residual network as IQ-ResNet. To accommodate the CM input, we modify some of the parameters in Fig. 3. The size of the convolution kernel in the first convolutional layer is 3×3 with a step size of $(1, 1)$. As for the maximum pooling layer, a convolutional kernel size of 3×3 is utilized, along with a stride of $(2, 2)$. In the four residual block structures, the two convolutional layers used for feature extraction have a convolutional kernel size of 2×2 and a step size of $(1, 1)$. In addition, the convolutional layer responsible for direct mapping has a convolutional kernel size of 1×1 and a step size of $(1, 1)$. For IQ-ResNet, we

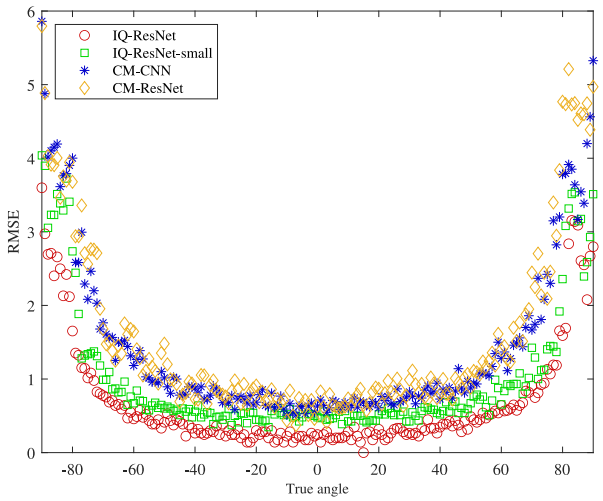


Fig. 17. Performance under different network structures.

also explore the performance of a smaller residual network model called IQ-ResNet-small. Compared with IQ-ResNet, we reduce the number of residual blocks and adjust some parameters to make the model much smaller. Specifically, the number of output channels for the first convolution layer is set to 16, and the output size of the largest pooling layer is one-eighth of the input size. The output channels of the two residual blocks are 32 and 64 respectively, and the output size is a quarter of the input size. The rest of the structure of the model remains unchanged.

In the single-source scenario, we use SNR $= -10$ dB samples (specifically, 100 test samples per DOA) to compare the performance of these methods. The experimental results are shown in Fig. 17. We observe that CM-ResNet and CM-CNN perform roughly the same in DOA estimation, both worse than the methods based on IQ input. Although the network structure changes, it does not improve the performance when using CM as the input. IQ-ResNet-small does not perform as well as IQ-ResNet, but still outperforms CM-CNN and CM-ResNet. These results indicate that IQ input is more advantageous for feature learning of deep neural networks in DOA estimation tasks compared to CM input.

2) Complexity Analysis: To analyze the time and space complexity of the methods, we need to consider the specific architecture and configuration of the models. By calculating the floating-point operations (FLOPs) and parameters (Params) of each model, we can determine their respective time and space complexity. The formula for calculating the FLOPs of a convolutional layer is:

$$\text{FLOPs} \sim O\Big(F^2 \cdot C_{in} \cdot C_{out}\Big), \tag{32}$$

where F is the length of each convolution kernel, C_{in} is the number of input channels, and C_{out} is the number of output channels. The space complexity encompasses the overall amount of weight parameters across all layers, commonly referred to as the "total number of parameters". The spatial complexity of a model is unrelated to the size of the input data and instead depends on factors such as the dimensions of the convolutional kernel, the number of channels, and the depth of the network.

TABLE I
FLOPs AND PARAMS OF DIFFERENT METHODS

Network \ Complexity	FLOPs	Params
IQ-ResNet-small	1.76729×10^6	4.6069×10^4
IQ-ResNet	8.71413×10^7	1.85285×10^6
CM-CNN	3.40764×10^7	2.09121×10^7
CM-ResNet	5.15018×10^7	2.37375×10^6
CM-MLP	$\mathbf{5.5724 \times 10^5}$	3.4055×10^4
SP-CNN	8.44613×10^5	$\mathbf{1.865 \times 10^3}$

For the method of CM-based methods, we need to consider the complexity of the computation of CM. When calculating the covariance matrix, each point needs to perform T times complex multiplication and $T-1$ times complex addition. Each complex multiplication involves four real multiplication operations and two real addition operations, while each complex addition requires two real addition operations. Therefore, the time complexity of covariance matrix computation is $M^2 \times (8T-2)$, space complexity is M^2.

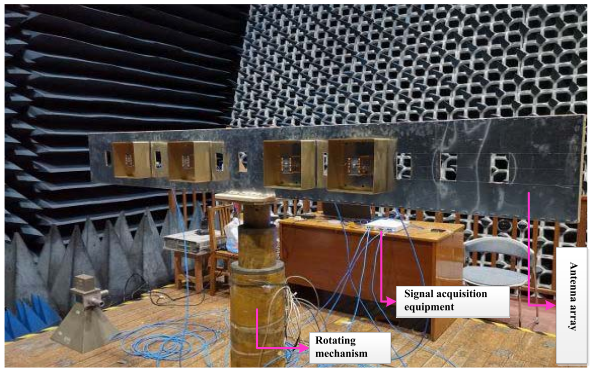
Here we consider the time complexity and space complexity for the single signal scenario. FLOPs and Params of our proposed method and other methods are shown in Table I. The CM-MLP method has the smallest time complexity. The SP-CNN method has minimal space complexity. IQ-ResNet-small is superior to CM-CNN in both time complexity and space complexity and also shows a greater advantage in performance. In terms of space complexity, our proposed IQ-ResNet method is lower than CM-CNN and CM-ResNet, and higher than CM-MLP and SP-CNN. These findings further validate the effectiveness of our proposed method.

VI. Experimental Results

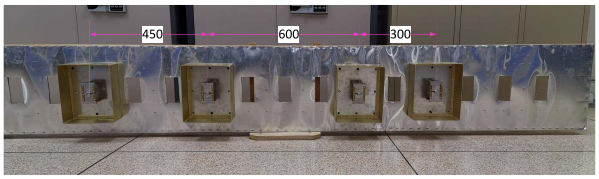
In the previous section, we have analyzed the performance of the proposed method through simulations. In this section, we further analyze the performance through the signals collected over-the-air in an anechoic chamber by the real-world test system.

A. Experiment Setup

We construct a practical test system comprising a signal source, rotating mechanism, a signal acquisition equipment, and an antenna array as shown in Fig. 18(a). The anechoic chamber can shield against various electromagnetic interference, which is more conducive to testing the performance of the methods. In the anechoic chamber, the antenna is located above the rotating mechanism which can rotate freely, and the signal source is located at a distance of 10 meters in front of the antenna. The receiver's antenna array is a linear array equipped with four antennas, operating within the frequency band of 960 MHz to 1250 MHz, with spacing between adjacent antennas set at 300 mm, 600 mm, and 450 mm respectively, as depicted in Fig. 18(b). The antenna incidence angle is configured to range from $-60°$ to $60°$,



(a)



(b)

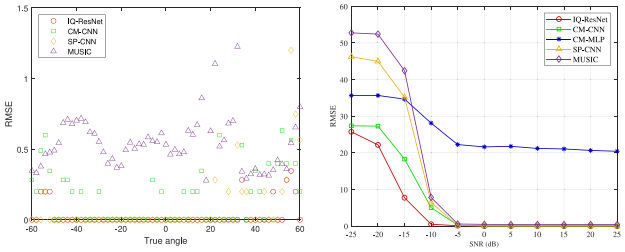
Fig. 18. The test system. (a) Experimental equipment and setup, (b) antenna array.

with a step size of 2°. The signal source transmits a QPSK-modulated signal at a power level of -15 dBm and a carrier frequency of 990 MHz. To simulate a noisy environment and accurately evaluate the model's robustness, we add artificial simulated noise to the received signal despite it being relatively noise-free due to the proximity of the transmitter and receiver. The SNR of the received signal after noise injection is varied from -25 dB to 25 dB in 5 dB intervals, yielding a total of eleven SNR levels. We used $T = 512$ snapshots. The training set is comprised of 100 samples in each direction at each SNR level, resulting in a total of 67,100 samples. The validation set is constructed similarly to the training set.

B. Experimental Results

We first select samples with SNR of -5 dB and snapshot number of $T = 512$ to validate the performance of our proposed method. A total of 100 test samples are chosen for each DOA. The experimental results are presented in Fig. 19(a). The IQ-ResNet method exhibits errors in only eight angles in the entire DOA region, with a maximum RMSE of 0.34. The CM-CNN method displays errors in 28 angles across the entire region, with a maximum RMSE of 0.63. The SP-CNN method shows that there are 12 angle errors in the whole area, and the maximum RMSE is 1.2. The RMSE of the MUSIC method is in the range [0.28, 1.22]. Since the RMSE of CM-MLP method is too large in this scenario, its result is not given here. Therefore, we can infer that the proposed IQ-ResNet method outperforms the other methods in this case, which further validate the superiority of the proposed method.

We next consider the performance in the SNR range of -25 dB to 25 dB with a 5 dB interval. There are 100 test samples for each angle at each SNR, resulting in a total of 6,100 test samples for each SNR. The results of the experiments are presented in Fig. 19(b). In the low SNR region, we observe a



(a) (b)

Fig. 19. DOA estimation performance of over-the-air collected signals. (a) Comparison of different DOA estimation methods, (b) performance under different SNRs.

decrease in the error of all five methods as the SNR increases. In the high SNR region, the RMSE of IQ-ResNet, CM-CNN SP-CNN and MUSIC methods is close to 0. Overall, the proposed method outperforms the other methods in terms of accuracy. These results highlight the effectiveness of the proposed method for DOA estimation in a wide range of SNR conditions.

VII. Conclusion

We have proposed a novel approach for DOA estimation of single and multiple signal sources using a residual network with IQ as input. We have presented two approaches for DOA estimation: a classifier-based method and a regression-based method. We have evaluated the performance of our method in different scenarios. Results have shown that our approach outperforms the existing deep learning-based DOA methods with CM as input in terms of accuracy and robustness. Furthermore, we have demonstrated the effectiveness of our approach with over-the-air acquired signals using our built test system, showing that our method can still accurately estimate the DOA in such conditions.

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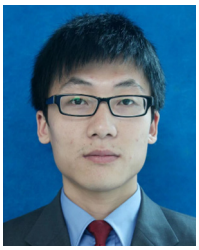

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